2023 11th International Symposium on Electronic Systems Devices and Computing (ESDC) | 978-1-6654-5572-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/ESDC56251.2023.10149861

UAV and UGV Assisted Path Planning for Sensor Data Collection in Precision Agriculture

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***Abstract*—In recent years, automated (Intelligent) decision support systems have become prevalent in various smart city applications such as healthcare, transportation, energy management, and environmental monitoring. Internet of Things (IoT) and Unmanned Aerial Vehicle (UAV)-based smart sensing and actuation devices in agricultural events can change the sector from static and manual to dynamic and intelligent, resulting in increased production with minimal human efforts. In addition, soil measurements that are time-consuming can be collected using Unmanned Ground Vehicles (UGVs). However, to collect data efficiently from wireless sensors in agricultural fields, UAV and UGV need to follow an optimal path. Thus, in this paper, we formulate utility maximization problem using UAV and UGV by simultaneously minimizing energy consumption and maximizing data collection. To solve the formulated problem, we propose a modified Greedy Randomised Adaptive Search Procedure (GRASP) algorithm to predict an efficient path for UAV and UGV to collect data from the agricultural field. Moreover, the efficacy of the proposed algorithm is showcased theoretically and experimentally on real-world data and compared with other state-of-the-art methods.**

***Index Terms*—UAV, UGV, Data collection, Agriculture, Sensors.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) and Unmanned Ground Vehicles (UGVs) can play a significant role in smart agriculture. By combining the two, farmers can benefit from increased efficiency, reduced labor costs, and improved crop yields [1]–[3]. UAVs can be used to gather data on crop health and growth by using the Agricultural Internet of Things (AIoT) sensors and cameras to capture images and other data. This information can then be processed to create maps of the fields, which can help farmers to identify areas of the crops that are not growing well, allowing for targeted interventions. UAVs can also be used to spray pesticides or fertilizers, reducing the need for manual labor and increasing the precision of the application. UGVs can be used to collect soil samples and perform other tasks such as irrigation and weed removal. By using UGVs, farmers can reduce the time and effort required to carry out these tasks and improve the accuracy and efficiency of their operations. Together, UAVs and UGVs can provide a comprehensive solution for smart agriculture. By collecting data on crop health and soil conditions, farmers can make informed decisions on using fertilizers, pesticides, and other interventions. This can help to reduce costs, increase crop yields, and improve the overall sustainability of agricultural practices.

A UAV or UGV (or a group of them) needs to visit various locations in the agricultural field to collect data from agricultural sensors [4]. Due to the limited storage capacity and data overwriting problem, the data from the sensors should be collected after a certain time interval [5]–[11]. However, efficient path prediction is required to minimize energy consumption during flight and hovering while collecting data from the sensors. Furthermore, certain parameters of soil measurements require significant amounts of time during their collection. Employing UGVs to cover these sensor areas is more efficient than using UAVs. Furthermore, after data collection, UAV and UGV return to the data center residing in the middle of the agricultural field for further processing.

To tackle the above-mentioned problem, we propose UAV and UGV-based data collection as utility maximization problems by minimizing the energy consumption of UAV and UGV as given in Fig. 1. Moreover, we show that the formulated problem is similar to the well-known NP-hard Orienteering Problem (OP). To solve the formulated problem in an efficient way, we propose a modified Greedy Randomised Adaptive Search Procedure (GRASP) [12] algorithm to predict the path for both UAV and UGV. The primary contributions of this paper are summarized in the following:

1) Formulate the problem of collecting data from agricultural field sensors using UAVs and UGVs as a utility maximization problem by minimizing energy consump-

tion while collecting data, as an NP-hard problem.

2) Solve the formulated problem efficiently using a modified GRASP algorithm by finding an efficient path for both UAV and UGV and maximizing the amount of data collection.
3) Through simulation study on real-world data, we have demonstrated that the proposed approach outperforms the state-of-the-art schemes.

The rest of the paper is organized as follows: Section II reviews relevant works. The system model and the problem formulation are introduced in Section III. The proposed solution and performance study are discussed in Sections IV and V, respectively. Section VI offers conclusions and future works.

II. Related Works

This section provides a comparison of closely related works in the literature, as shown in Table I. Authors of [4], proposed a utility maximization problem to maximize data collection by optimizing UAV's path. The authors introduced a utility function and maximized the utility through various potential hovering locations to get an optimized trajectory for the UAV. However, they did not consider the deployment of UGV, focusing solely on the UAV's trajectory. The authors of [12] discussed Close Enough OP (CEOP), which focused on finding the most rewarding route visiting different regions based on the given travel budget using the GRASP algorithm. However, they did not take into account utility maximization for UAV and UGV models. The authors of [13], described a sensor planning scheme based on the path of the UGV by minimizing a cost function. The authors showed that UGV carries UAV to different points, and then UAV collects data. However, the authors did not consider the utility maximization problem for the trajectory. The authors of [14], conducted an analytical study that examined the proximity at which a drone equipped with a LoRa radio had to fly over sensors in order to achieve a specific level of data collection quality. However, the authors did not consider path prediction and utility maximization in their proposed model. The authors of [15], formulated OP for curvature-constrained vehicles. They proposed the Variable Neighborhood Search (VNS) technique and attempted to maximize rewards while ensuring the tour length adhered to a specified travel budget restriction.

TABLE I: Summary of related works

Problem Focus	Path Prediction	Utility Maximization	UAV + UGV	LoRa
Utility Maximization [4]	✓	✓	×	×
GRASP [12]	✓	×	×	×
Sensor Planning [13]	✓	×	✓	×
Precision Agriculture [14]	×	×	×	✓
VNS [15]	✓	✓	×	×
Proposed approach	✓	✓	✓	✓

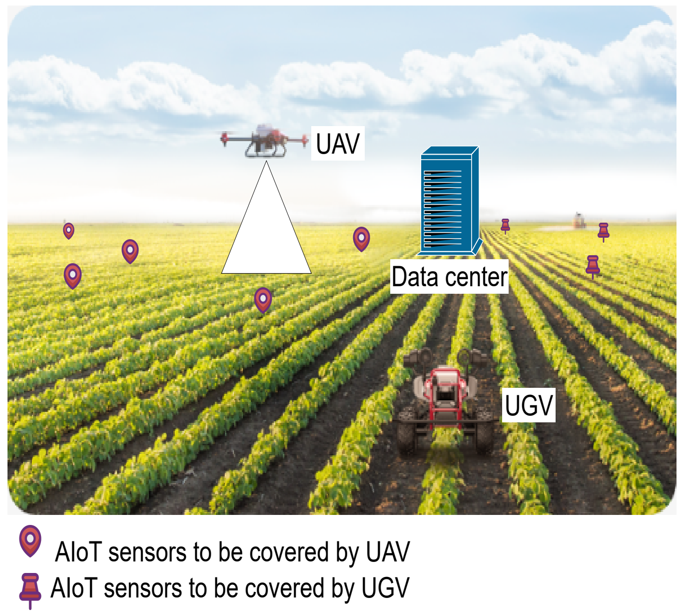

Fig. 1: System model.

III. System Model and Problem Formulation

We consider a combined UAV and UGV model with MEC server as a data center for agricultural data collection, as shown in Fig. 1. The data transmission from AIoT sensors to UAV or UGV uses LoRa transceivers. Let $\mathbb{S} = \{1, \cdots, s, \cdots, S\}$, $\mathbb{G} = \{1, \cdots, g, \cdots, G\}$ and $\mathbb{A} = \{1, \cdots, a, \cdots, A\}$ be the sets of all AIoT sensors in the field, AIoT sensors to be covered by the UGV, and the AIoT sensors to be covered by the UAV, respectively. UAV and UGV cover different parts of the agricultural field to collect data from AIoT sensors by finding near-optimal paths. Both UAV and UGV start from the data center located in the middle of the field. After collecting data, UAV and UGV return to the data center on each tour. A closed data collection tour of the UAV starts and ends at the data center, consisting of a sequence of hovering locations at altitude h, visited by the UAV to collect agricultural data. However, it is not feasible to collect all the data in a single tour due to energy constraints. Thus, multiple tours are conducted by the UAV and UGV to gather data from the AIoT sensors located throughout the fields[1].

A. Path Planning

Assuming that the data center is situated at the center of the agricultural field, the data collection process is done by deploying UAV and UGV for collecting data from the sensors, which can be described as follows:

1) We assume a random path as the initial path.
2) Other hovering locations are injected into the path until the cost is under the budget.

[1]As the quantity of orchards requiring data management expands, there is a proportional increase in both the number of data centers needed to handle the workload and the frequency of UAV and UGV tours.

3) Once a tour is found, a local search algorithm gets initiated to ensure a near-optimal tour.
4) Once a near-optimal tour has been obtained, all the locations in the set of hovering locations that are not part of this tour are removed.

B. Data Collection Tour Model

In the proposed model, UAV and UGV start from the data center and return to the data center after collecting data from AIoT sensors in each tour.

1) UAV: A tour of the UAV consists of a sequence of hovering locations at an altitude of h above the ground. Let $\mathbb{H}$ be the set of $\mathfrak{H}$ hovering locations as given by:

$$\mathbb{H} = \{(x_1, y_1, h), \cdots, (x_a, y_a, h), \cdots, (x_{\mathfrak{H}}, y_{\mathfrak{H}}, h)\}, \tag{1}$$

where (x_a, y_a, h) is the hovering coordinate for the UAV to collect data from AIoT sensor a. Then, the total flight distance f_d for the UAV with total flight duration Γ, travelling with the speed of v_{UAV} in a tour is given as follows:

$$f_d = v_{UAV} \times \Gamma. \tag{2}$$

Thus, the UAV can travel to a hovering location (x_a, y_a, h) and back to the data center only if $x_a^2 + y_a^2 + h^2 \leq R^2$, where $R = \frac{f_d}{2}$. Let $t_d = \frac{D_a}{B}$ be the time taken by the UAV to collect data from a^{th} sensor, where D_a is the data size at sensor a, and B is the channel bandwidth [14]. Then, the hovering time Ψ_a for each hovering location is given below:

$$\Psi_a = min(t_d, t_{DC}), \tag{3}$$

where t_{DC} is the duty cycle of LoRa trans-receiver [14].

The distance that the UAV travels from its current location to the next hovering location is given as follows:

$$d_{a',a} = \sqrt{(x_a - x_{a'})^2 + (y_a - y_{a'})^2}, \tag{4}$$

where (x_a, y_a) and $(x_{a'}, y_{a'})$ are the next and the current hovering coordinates for UAV. The time required for the UAV to travel and collect data from sensor a is calculated as follows:

$$T_a = \frac{d_{a',a}}{v_{UAV}} + \Psi_a. \tag{5}$$

However, due to energy constraints, UAV cannot collect data from all sensors during a single tour. Therefore, the UAV should prioritize more important hovering locations over others.

2) UGV: A UGV tour consists of a series of locations where transmissions occur. The set $\mathbb{N}$ contains coordinates of UGV for collecting data from sensors in $\mathbb{G}$, as follows:

$$\mathbb{N} = \{(\mathfrak{x}_1, \mathfrak{y}_1), \cdots, (\mathfrak{x}_g, \mathfrak{y}_g), \cdots, (\mathfrak{x}_N, \mathfrak{y}_N)\}, \tag{6}$$

where $(\mathfrak{x}_g, \mathfrak{y}_g)$ is the coordinate corresponds to AIoT sensor g.

Assuming v_{UGV} is the velocity of the UGV, the distance between the current location $(\mathfrak{x}_{g'}, \mathfrak{y}_{g'})$ and the next location $(\mathfrak{x}_g, \mathfrak{y}_g)$ is calculated as follows:

$$d_{g',g} = |(\mathfrak{x}_g - \mathfrak{x}_{g'})| + |(\mathfrak{y}_g - \mathfrak{y}_{g'})|. \tag{7}$$

Suppose Ψ_g^{samp} represents the time needed to sample the data at $(\mathfrak{x}_g, \mathfrak{y}_g)$. Then, Ψ_g^{samp} is calculated as [14]:

$$\Psi_g^{samp} = \frac{D_g}{B}, \tag{8}$$

where D_g denotes the data size collected by sensor g.

C. Data Collection Utility Model

The quantity of data gathered by a sensor on a UAV or UGV is directly linked to the duration of the hovering or transmission associated with the sensor. We express the amount of data collected from sensor s during a period of time t using a function $F_s(t)$ as follows [4]:

1) $F_s(t) = 0$ if $t = 0; F_s(t) > 0$ if $0 < t \leq \Psi$; otherwise $F_s(t) = F_s(\Psi)$ if $t > \Psi$.
2) Function $F_s(t)$ is non-decreasing, i.e. $F_s(t_1) \leq F_s(t_2)$ for $0 \leq t_1 < t_2$.
3) $F_s(t)$ is a sub modular function, where $F_s(t_1 + \Delta) - F_s(t_1) > F_s(t_2 + \Delta) - F_s(t_2)$ for $0 \leq t_1 < t_2 \leq \Psi$ and $F_s(t_1 + \Delta) - F_s(t_1) = F_s(t_2 + \Delta) - F_s(t_2)$ for $\Psi \leq t_1 < t_2$.

where $\Psi = \Psi_a$ when UAV is considered and $\Psi = \Psi_g^{samp}$ while assuming UGV.

1) UAV: Let the priority of all hovering locations in $\mathbb{H}$ is given by the set $\mathbb{P}_{UAV} = \{P_1, \cdots, P_a, \cdots, P_{\mathfrak{H}}\}$, where P_a is the priority of the hovering location (x_a, y_a, h). Next, we define utility u_a of the UAV at hovering location (x_a, y_a, h) as follows:

$$u_a = F_a(\Psi_a) \times P_a. \tag{9}$$

Hence, the overall utility of the UAV is expressed as follows:

$$U_{UAV} = \sum_{(x_a, y_a, h) \in \mathbb{H}} u_a. \tag{10}$$

Let the set of potential UAV tours is denoted by $\boldsymbol{\tau} = \{\tau_0, \cdots, \tau_m, \cdots, \tau_{\mathcal{M}}\}$, and the energy consumption of the UAV per unit time during tour τ_m be ΔE_{UAV}^m. The cost of collecting data from sensor a on tour τ_m by UAV is given as follows:

$$c_{a,m} = \Delta E_{UAV}^m \times T_a. \tag{11}$$

Thus, the total cost of UAV for travelling to each hovering location in the tour is defined as follows:

$$C_{UAV} = \sum_{(x_a, y_a, h) \in \tau_m, \tau_m \in \boldsymbol{\tau}} c_{a,m}. \tag{12}$$

2) UGV: Let $\mathbb{P}_{UGV} = \{\mathcal{P}_0, \cdots, \mathcal{P}_g, \cdots, \mathcal{P}_{\mathcal{H}}\}$, where $\mathcal{P}_g$ is the priority of transmission location at $(\mathfrak{x}_g, \mathfrak{y}_g)$. We define the utility of the UGV at transmission location $(\mathfrak{x}_g, \mathfrak{y}_g)$ as follows:

$$u_g = F_g(\Psi_g) \times \mathcal{P}_g. \tag{13}$$

Thus, the total utility of UGV is calculated as follows:

$$U_{UGV} = \sum_{(\mathfrak{x}_g, \mathfrak{y}_g) \in \mathbb{N}} u_g. \tag{14}$$

Let ΔE_1 and ΔE_2 be the cost of travelling energy and transmission energy per unit time. As a result, the total energy consumed by the UGV while travelling to and collecting data from the sensor at $(\mathfrak{x}_g, \mathfrak{y}_g)$ is as follows:

$$E_g = \Delta E_1 \times \frac{d_{g',g}}{v_{UGV}} + \Delta E_2 \times \Psi_g^{samp}. \tag{15}$$

Therefore, the cost of UGV for travelling and collecting data from the sensors is defined as follows:

$$C_{UGV} = \sum_{(\mathfrak{x}_g,\mathfrak{y}_g)\in\mathbb{N}} E_g. \tag{16}$$

D. Problem Statement

Formally, the utility maximization problem while minimizing energy consumption is formulated as follows:

$$\textbf{P:}\ \ \frac{max\ (\sum_{(x_a,y_a,h)\in\tau_m} u_a + \sum_{(\mathfrak{x}_g,\mathfrak{y}_g)\in\mathbb{N}} u_g)}{min\ \ (C_{UAV} + C_{UGV})} \tag{17}$$

s.t.

$$\sum_{a\in\tau_m} T_a \le \Gamma \tag{17a}$$

$$0 < C_{UAV} \le Th_{UAV} \tag{17b}$$

$$0 < C_{UGV} \le Th_{UGV} \tag{17c}$$

Constraint 17a states that the total time for a tour must be less than the budget time, constraint (17b) states that total accumulation cost for UAV should not exceed the threshold limit of Th_{UAV}, constraint (17c) represents that total accumulation energy should be less than Th_{UGV}.

E. Formulation as a Graph Problem

The formulated problem can be considered a graph problem as described below:

1) UAV: Let $\mathcal{G}_{UAV}$ be a graph with vertex set $V_{UAV} = \mathbb{H}$, representing the hovering locations of the UAV, and edge set E_{UAV} defined as follows:

$$E_{UAV} = \{(\mathfrak{r}, \mathfrak{s}) : \forall \mathfrak{r}, \mathfrak{s} \in V_{UAV}\}. \tag{18}$$

Cost of the edge between vertices $\mathfrak{r}$ and $\mathfrak{s}$ is given as follows:

$$c_{\mathfrak{r},\mathfrak{s}} = \frac{\psi_\mathfrak{r} + \psi_\mathfrak{s}}{2} + \frac{d_{\mathfrak{r},\mathfrak{s}}}{v_{UAV}}. \tag{19}$$

where, $\psi_\mathfrak{r}$ and $\psi_\mathfrak{s}$ is the hovering time at $\mathfrak{r}^{th}$ and $\mathfrak{s}^{th}$ vertex respectively.
We consider the utility u_a of a^{th} sensor as the reward for visiting a^{th} vertex in the graph.

2) UGV: Let $\mathcal{G}_{UGV} = (V_{UGV}, E_{UGV})$ be a graph, where V_{UGV} denotes the set of possible $\mathbb{N}$ transmission locations of the UGV, and E_{UGV} is defined as follows:

$$E_{UGV} = \{(r, s) : \forall r, s \in V_{UGV}\}. \tag{20}$$

Cost of the edge between vertices r and s is given as follows:

$$c_{r,s} = \frac{\psi_r^{samp} + \psi_s^{samp}}{2} + \frac{d_{r,s}}{v_{UGV}}, \tag{21}$$

where, ψ_r^{samp} and ψ_s^{samp} is the sampling time at r^{th} and s^{th} vertex respectively.
We consider the utility u_g of g^{th} sensor as the reward for visiting g^{th} vertex in the graph.

Theorem 1. *Utility maximization problem **P** through path-finding is NP-hard.*

Proof. Well-known OP [16] can be reduced to the proposed formulated problem.

We define OP using a graph $\mathcal{G}_{OP} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \nu_i : i = 1, 2, \cdots, \mathcal{N}$ is the set of vertices and $\mathcal{E}$ is the set of arcs. In OP, each vertex $\nu_i \in \mathcal{V}$ of a graph $\mathcal{G}_{OP} = (\mathcal{V}, \mathcal{E})$ is associated with a non-negative score S_i, and each arc $e_{i,j} \in \mathcal{E}$ has an associated cost $c_{i,j}$. The goal of OP is to find a Hamiltonian path from vertex ν_1 to vertex $\nu_\mathcal{N}$ such that the total cost of the edges in the path does not exceed a maximum limit C_{max} and the total score collected along the path is maximized [16]. We map the OP to the formulated problem **P** by considering the starting and ending point of the path as the location of the data center, as follows:

- Map the vertex set $\mathcal{V}$ to $\mathbb{H}$ vertices of AIoT sensors for UAV and $\mathbb{N}$ vertices of AIoT sensors for UGV.
- Map the edge set $\mathcal{E}$ to the edges of $\mathcal{G}_{UAV}$ and $\mathcal{G}_{UGV}$
- Map the reward in OP to the utility.
- Map maximum cost limit to the travel budget.

TABLE II: OP & Path finding

OP	**Path finding**
Vertex set $\mathcal{V}$	Set of sensor locations
Edge set $\mathcal{E}$	E_{UAV}, E_{UGV}
Reward in OP	Utility
Maximum cost limit C_{max}	Time budget

Table II shows the mapping of the path-finding to the OP [16], which is NP-hard, implying the formulated utility maximization problem **P** is an NP-hard. □

IV. Proposed Solution

Algorithm 1, demonstrates a path selection for UAV and UGV to collect sensor data. Lines 1 and 2, calculate utility for i^{th} vertex. Lines 3 to 4 calculate the utility value for each vertex in the vertex set $\mathcal{V}$. Lines 5 to 6 calculate the edge cost for each edge in the edge $\mathcal{E}$ set. An auxiliary graph is constructed with a dummy data center d_o. The GRASP algorithm returns a path and sequence of visits. A near-optimal closed tour is then derived from the returned path.

V. Performance Study

In this section, we demonstrate the effectiveness of the proposed model through simulation analysis. The analysis uses data obtained from the United States (US) agriculture department website, which includes information on various parameters such as the number of sensors, depth of measurement, locations of field sensors, and other parameters [17]. We have conducted a comparative analysis of the results of

Algorithm 1 Proposed Path Finding Algorithm

Input: A graph $\mathcal{G}_s = (\mathcal{V}, \mathcal{E})$, a device with energy capability ξ at, and each node i $\in \mathcal{V}$ has preference P_i.
Output: A closed tour τ derived from path ρ.

for $i \in \mathcal{V}$
    Calculate utility using Eq. (9) and Eq. (13)
for $(i, j) \in \mathcal{E}$
    Calculate $c_{i,j}$ cost using Eq.(19) and Eq. (21)
Construct a graph $\mathcal{G}_s = (\mathcal{V} \cup \{d_o\}, \mathcal{E} \cup \{(v, d_o)|(v, d) \in \mathcal{E}\})$; edge cost $c_{d_o,d} = 0$, where d_o is a dummy data center
Find a path ρ in G_s between the data center d and dummy data center d_o such that the total utility of vertices visited in the path is maximized, subject to total cost constraints Th_{UAV} and Th_{UGV} of the UAV and UGV and given time budget by the GRASP Algorithm [12]
return: A closed tour τ, is obtained from the path ρ.

the proposed model with the existing works, VNS [15] and GRASP [12]. VNS uses a combinatorial heuristic approach to insert, or exchange paths, and GRASP uses a combinatorial metaheuristic technique to solve the addressed CEOP. Moreover, the values taken for different parameters are shown using Table III.

TABLE III: Simulation parameters

Parameter	Values
Sensor data size	20-238.5 KB [18]
Number of sensors	10 [19]
UAV and UGV	1,1
Th_{UAV}, Th_{UGV}	1000, 1500
Γ	50
P_a	1.1 mW [20]

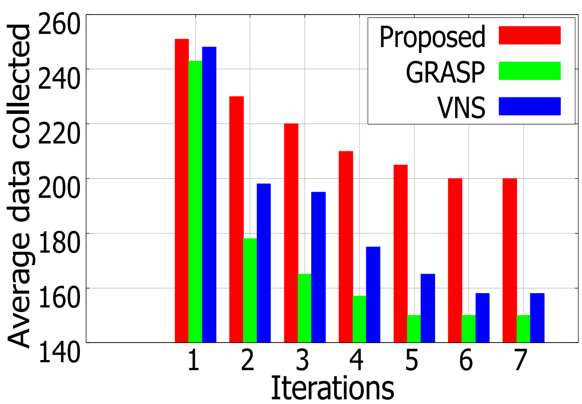

Fig. 2: Average data collection analysis.

In Fig. 2, we have plotted the average data collected in the tours taken by the UAV and UGV in the proposed model, and the data collected by UAV in the case of VNS and GRASP. We observe that the average amount of data collected by using the proposed model is higher as compared to that of VNS and GRASP. The main reason is that UAV is battery constrained device, so it covers fewer sensors when compared with UAV and UGV in the proposed model together.

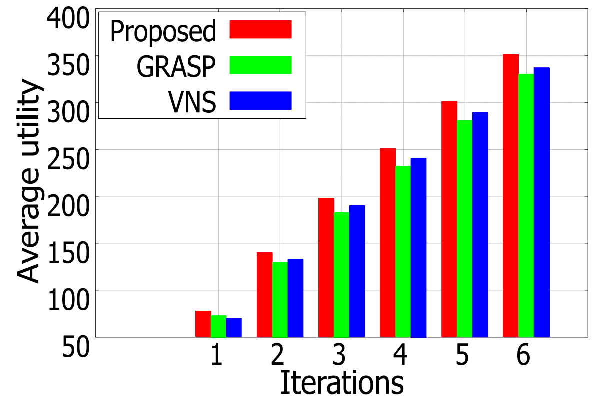

Fig. 3: Average utility analysis.

Fig. 3, shows the average utility against the number of iterations of the proposed model, VNS, and GRASP. We have plotted the average utility of the tours taken by UAV and UGV in the proposed model, and UAV in the case of VNS and GRASP. We observe that the utility of the proposed model is generally more as compared to that of VNS and GRASP. The reason is that the proposed model finds the best path for UAV and UGV to maximize data collection from all sensors in the field with less energy consumption.

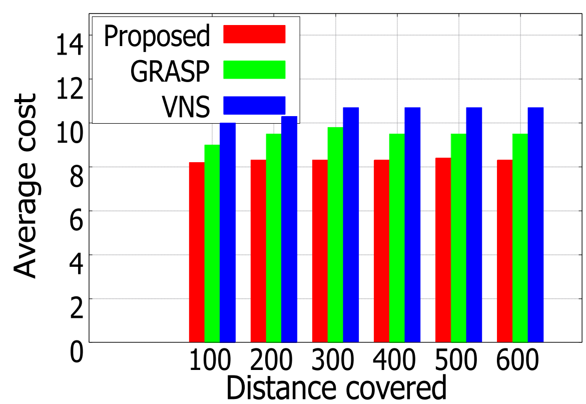

Fig. 4: Average cost v/s distance.

Fig. 4, compares the average cost of the proposed model, VNS and GRASP. The average cost of the tours taken by the UAV and UGV is plotted against the distance covered by them. We observe that the cost of the proposed model is less compared to that of VNS and GRASP. The reason is that the proposed model finds the near-optimal path of UAV and UGV to gather data from sensors with less energy consumption.

Fig. 5, compares the total utility of the proposed model, VNS and GRASP. We have plotted the total utility of all the

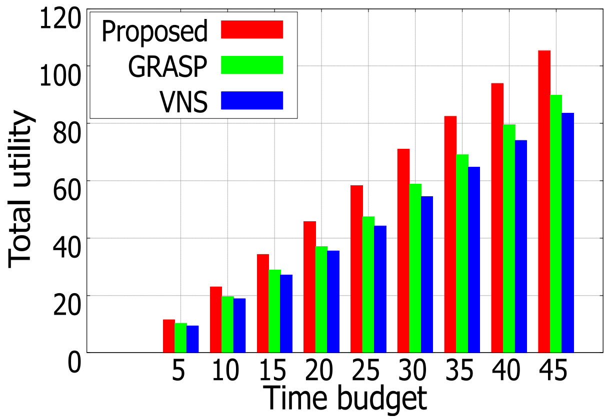

Fig. 5: Total utility v/s time budget.

tours taken by the UAV and UGV against the budget provided. Since the VNS and GRASP did not consider UGV for data collection, the curve of VNS and GRASP is plotted based on the UAV only. We observe that the utility obtained by the proposed model is more as compared to that of VNS and GRASP. The reason is that the proposed model used both UAV and UGV for collecting the data, which gives better results over models with only UAV collecting the data and UGV used as a transport.

VI. Conclusion

In this work, we have formulated a total utility maximization problem considering energy cost into deliberation as NP-hard. In order to solve the formulated problem, we have utilized modified GRASP algorithm to predict path for UAV and UGV for collecting data from sensors in an agricultural field. Moreover, through simulation analysis on real-world data, we have shown the effectiveness of the proposed algorithmic model and compared it with other state-of-the-art methods. Additionally, the proposed model has applications beyond precision farming, as it can be utilized to remotely monitor patients in hilly or terrain areas by employing UAV and UGV to gather data from patients attached with Wireless Body Area Network (WBAN) sensors.

In future work, we aim to employ suitable Machine Learning techniques to analyze the collected data for predicting the health and environmental conditions of the crops in real-time.

Acknowledgment

This work is supported by the Science and Engineering Research Board (SERB), Government of India, under Grant SRG/2020/000318.

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